EE1060: Differential Equations and Transform Techniques

Practice Problem Set - 10

Practice problems in topics covered till: Lecture 36

Note : The practice problem sets are designed for your personal review and are not required for submission or evaluation. These problems aim to reinforce your understanding of the material and help you prepare effectively. Some of the practice problems are sourced from the course references, and it is strongly recommended that you consult these references for additional practice problems. Due credit is also given to the TAs of the course, who have contributed to the creation of this problem set.

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1. Consider the second-order ODE
$$\frac{d^2x}{dt^2} + 2\zeta\omega_n \frac{dx}{dt} + \omega_n^2 \, x = 0 \tag{1}$$
   (a) Convert this second-order ODE into a system of first-order differential equations.

   (b) For $\omega_n = 10$, compute the eigenvalues and eigenvectors of the system matrix $\mathbf{A}$ for the following values of the damping ratio ζ:
      1. $\zeta = 0$
      2. $\zeta = 0.5$
      3. $\zeta = \frac{1}{\sqrt{2}}$
      4. $\zeta = 1$
      5. $\zeta = \sqrt{2}$

   (c) For each case above, describe or sketch the trajectories of the system's solution in the phase plane (i.e., the (x, x') plane).

2. Consider the system
$$\dot{\mathbf{x}} = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix} \mathbf{x} \tag{2}$$
   Verify if the following vectors are solutions.
$$\mathbf{x}_1(t) = \begin{bmatrix} e^t \\ 2e^t \end{bmatrix}, \quad \mathbf{x}_2(t) = \begin{bmatrix} e^{3t} \\ 4e^{3t} \end{bmatrix} \tag{3}$$
   In addition, determine their linear independence and sketch the phase plane trajectories.

3. Consider the system
$$\dot{\mathbf{x}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 3 \end{bmatrix} \mathbf{x} \tag{4}$$

Verify if the following vectors (not common) are solutions.

$$\mathbf{x}_1(t) = \begin{bmatrix} e^{t} \\ e^{-2t} \\ e^{3t} \end{bmatrix}, \quad \mathbf{x}_2(t) = \begin{bmatrix} e^{-t} \\ e^{2t} \\ e^{-3t} \end{bmatrix}, \quad \mathbf{x}_3(t) = \begin{bmatrix} e^{2t} \\ e^{-t} \\ e^{4t} \end{bmatrix} \tag{5}$$

In addition, determine the linear independence of these vectors and describe the trajectories in $\mathbb{R}^3$.

4. Consider the system

$$\dot{\mathbf{x}} = \begin{bmatrix} \sin(t) & 0 \\ 0 & \cos(t) \end{bmatrix} \mathbf{x} \tag{6}$$

Verify if the following vectors are solutions.

$$\mathbf{x}_1(t) = \begin{bmatrix} \cos(t) \\ \sin(t) \end{bmatrix}, \quad \mathbf{x}_2(t) = \begin{bmatrix} \sin(t) \\ -\cos(t) \end{bmatrix} \tag{7}$$

In addition, determine if they are linearly independent and sketch the phase plane trajectories.

5. Consider the system

$$\dot{\mathbf{x}} = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \mathbf{u} \tag{8}$$

Determine the conditions under which this system reduces to a homogeneous ODE.